

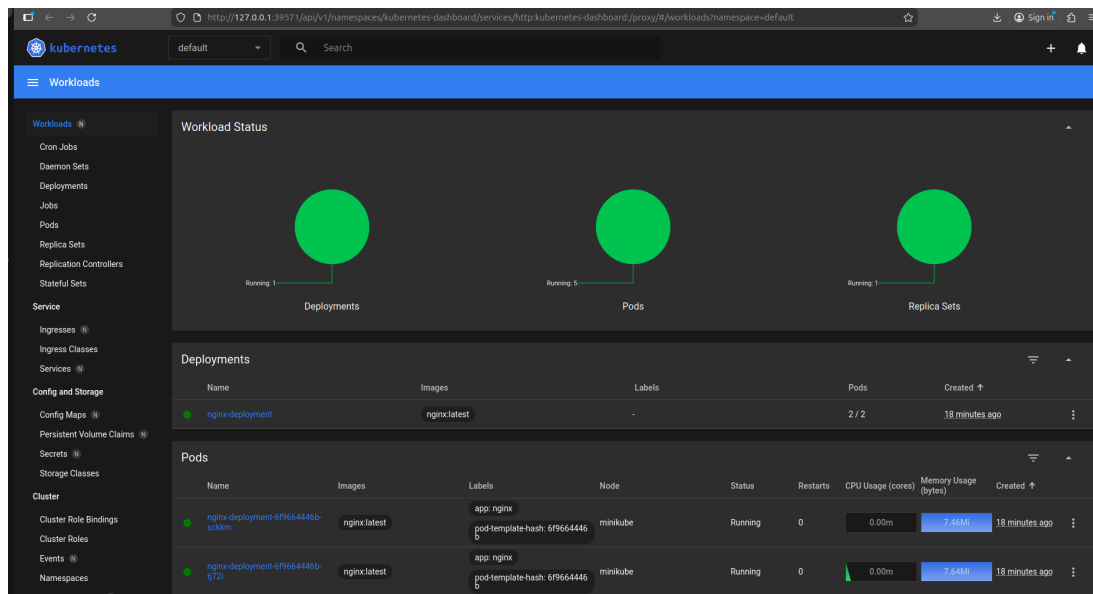
Program-10 Kubernetes Dashboard and CLI for Cluster Monitoring

Step 1: Start Minikube

minikube start --cpus=2 --memory=4096

Step 2: Enable Kubernetes Dashboard

minikube dashboard



Step 3: Create Deployment (deployment.yaml)

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Ga
GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 4: Create Service (service.yaml)

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-G...
GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

Step 5: Apply Resources and monitor via CLI

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

kubectl get pods

kubectl get nodes

kubectl get svc

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/prog10$ kubectl apply -f deploy
kubectl apply -f service.yaml
deployment.apps/nginx-deployment unchanged
service/nginx-service created
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/prog10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-pod                            1/1     Running   1 (6h48m ago)  7h32m
labeled-pod                         1/1     Running   1 (6h48m ago)  7h9m
nginx-deployment-6f9664446b-sckkm   1/1     Running   0           22m
nginx-deployment-6f9664446b-tj72l   1/1     Running   0           22m
simple-pod                           1/1     Running   1 (6h48m ago)  7h16m
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/prog10$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  35d   v1.34.0
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/prog10$ kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1        <none>       443/TCP          19d
nginx-deployment     NodePort    10.97.77.2       <none>       80:30823/TCP     21m
nginx-service        NodePort    10.109.250.84    <none>       80:32286/TCP     22s
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/prog10$
```

Step 6: Access Application

minikube service nginx-service

Opens NGINX welcome page in browser.

